

LOCUS ARCA

Laboratory Sample Management System User Manual

Version 1.0 · 2026

1. Introduction

Locus Arca is a browser-based application for managing laboratory samples. It runs entirely locally in your web browser —no installation, no server, no internet connection required. No admin rights are needed. All data is stored in a single JSON file on your computer.

1.1 Feature Overview

- Freezer unit management with optional location (room number, building, external provider)
- Visual groups for freezer units (e.g. "-20°C devices") —display only, not part of the sample path
- Hierarchical storage structure: Unit → Compartment → Rack → Drawer → Box —not all levels required
- Samples can be stored without a box, or without any storage location
- Three box types: Grid (A1, B2...), Numbered (1, 2, 3...), Freeform (text note)
- Grid boxes: visual position picker, occupied slots marked, no double-booking
- Multi-position selection in grid boxes (Ctrl+Click) —creates one sample per position
- Sample ID is optional —system auto-assigns internal ID (S-00001...) if left blank
- Multi-select with bulk actions: edit, move, delete
- Right-click context menu: rename, resize, move, delete storage elements
- Drag & drop reordering in the sidebar tree
- CSV export with selectable columns; CSV import with auto-creation of missing structures
- JSON storage with backup function and overwrite confirmation
- Full audit log
- German / English toggle

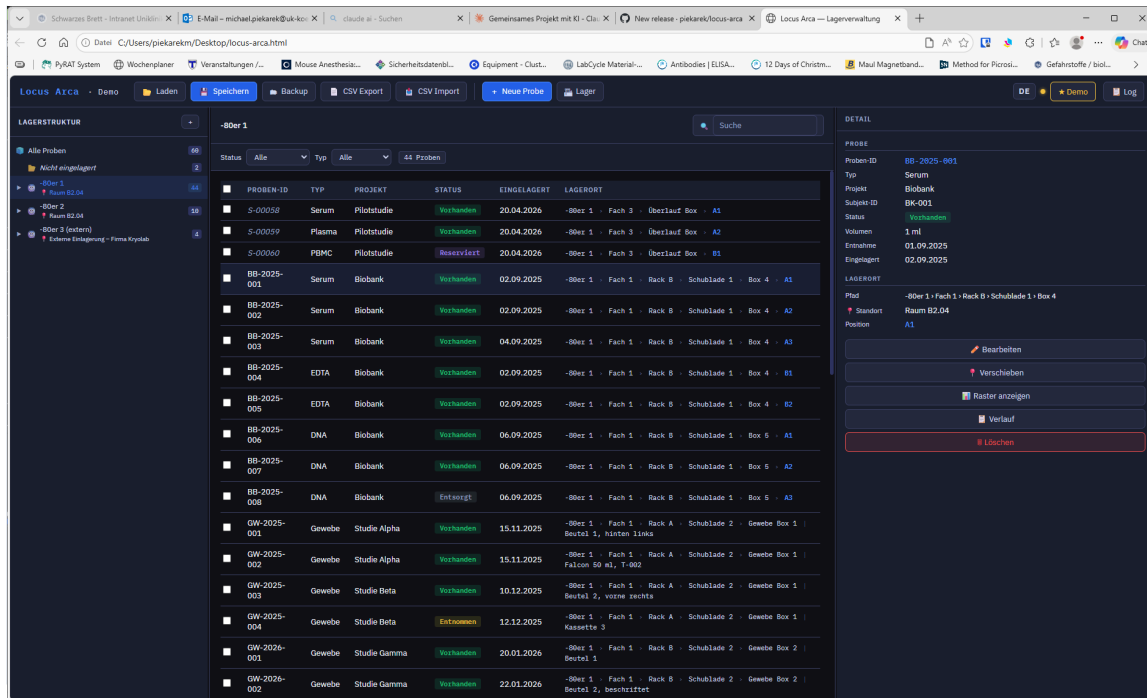
1.2 System Requirements

- Modern web browser —Chrome or Edge recommended (direct save); Firefox works with download fallback
- No installation required —the app is a single HTML file
- No internet connection needed · No admin rights required

■ Opera, Brave and Vivaldi also support direct saving as they share the same engine as Chrome. Safari and Firefox download the file instead.

2. User Interface

The interface consists of four areas: toolbar (top), sidebar (left), sample list (center), and detail panel (right).



2.1 Toolbar

- Load —opens an existing JSON database
- Save —saves to the current file; shows a confirmation prompt before overwriting
- Backup —always opens a Save As dialog
- CSV Export —opens field selection, then exports as CSV
- + New Sample —opens the sample form
- Storage —opens the storage structure manager
- DE / EN —language toggle
- Demo —loads example data
- Log —shows the audit log

The current filename is shown next to "Locus Arca" in the title bar.

2.2 Sidebar

The sidebar shows the storage structure as a tree. Right-click any item to edit, move or delete it. Drag and drop to reorder within the same level. Freezer units show their location below the name.

Freezer units can be organised into visual groups (e.g. "-20°C devices"). Groups are display only—they do not appear in sample storage paths.

A special entry appears automatically when relevant:

- ■ Not stored —samples with no storage location assigned



2.3 Sample List

Lists all filtered samples. Columns are sortable. Samples without a custom ID are shown in grey italics with their system ID (S-00001...). Checkboxes on the left allow multi-select; a bulk action bar appears when samples are selected.

All samples

Search

Status: All

Type: Alle

59 samples

3 samples selected

Edit selection

Move selection

Delete selection

X Deselect all

☐	SAMPLE ID	TYPE	PROJECT	STATUS	STORED ON	LOCATION
☒	S-00058	Serum	Pilotstudie	Available	20.04.2026	-80er 1 - Fach 3 - Überlauf Box - A1
☒	S-00059	Plasma	Pilotstudie	Available	20.04.2026	-80er 1 - Fach 3 - Überlauf Box - A2
☒	S-00060	PBMC	Pilotstudie	Reserved	20.04.2026	-80er 1 - Fach 3 - Überlauf Box - B1
☐	BB-2025-001	Serum	Biobank	Available	02.09.2025	-80er 1 - Fach 1 - Rack B - Schublade 1 - Box 4 - A1
☐	BB-2025-002	Serum	Biobank	Available	02.09.2025	-80er 1 - Fach 1 - Rack B - Schublade 1 - Box 4 - A2
☐	BB-2025-003	Serum	Biobank	Available	04.09.2025	-80er 1 - Fach 1 - Rack B - Schublade 1 - Box 4 - A3
☐	BB-2025-004	EDTA	Biobank	Available	02.09.2025	-80er 1 - Fach 1 - Rack B - Schublade 1 - Box 4 - B1
☐	BB-2025-005	EDTA	Biobank	Available	02.09.2025	-80er 1 - Fach 1 - Rack B - Schublade 1 - Box 4 - B2
☐	BB-2025-006	DNA	Biobank	Available	06.09.2025	-80er 1 - Fach 1 - Rack B - Schublade 1 - Box 5 - A1
☐	BB-2025-007	DNA	Biobank	Available	06.09.2025	-80er 1 - Fach 1 - Rack B - Schublade 1 - Box 5 - A2
☐	BB-2025-008	DNA	Biobank	Disposed	06.09.2025	-80er 1 - Fach 1 - Rack B - Schublade 1 - Box 5 - A3
☐	EX-2026-001	Serum	Kooperation XY	Available	11.02.2026	-80er 3 (extern) - Externe Box 1 - A1
☐	EX-2026-002	Serum	Kooperation XY	Available	11.02.2026	-80er 3 (extern) - Externe Box 1 - A2

2.4 Detail Panel

Shows all fields of the selected sample including full storage path, freezer location, box position, and storage note. Edit, Move, History and Delete buttons are available.

3. Getting Started

3.1 Opening the Application

Open locus-arca.html in Chrome or Edge. On first launch the database is empty. Use Load to open an existing JSON file.

- Place the JSON file on a shared network drive so all team members can access it. Only one person should edit at a time —simultaneous access can cause data loss.

3.2 Demo Data

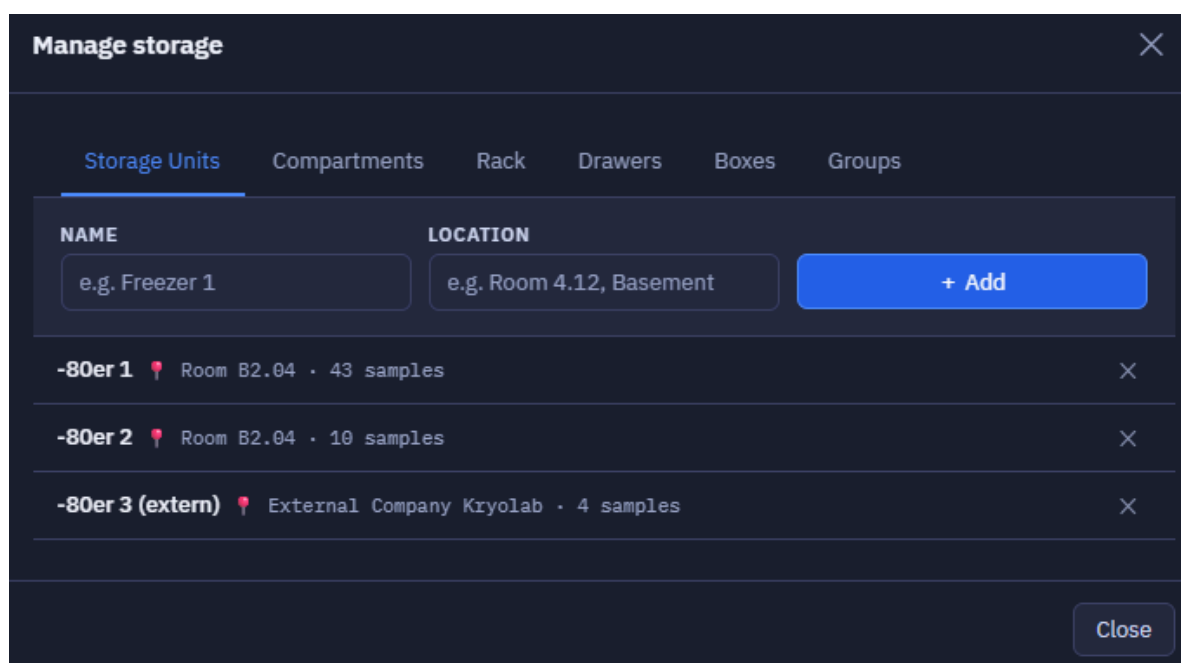
Click Demo in the toolbar to load realistic example data covering all features: samples without a box, samples without a storage location, samples without an ID, and various box types. Demo data can be discarded at any time.

3.3 Setting Up Storage Structure

Open the dialog via Storage. All levels are optional:

- Storage unit —name + optional location (room, building, external provider)
- Groups —visual folders for storage units; assign units via the Groups tab
- Compartment, Rack, Drawer —name only
- Box —name + type: Grid (fixed raster), Numbered (1...100), or Freeform (text note)

- A box can be assigned directly to a storage unit or compartment —rack and drawer are not required.



4. Managing Samples

4.1 New Sample

Click + New Sample or press Ctrl+N.

Field	Description
Sample ID	Optional display name. If empty the system assigns an ID (S-00001...)
Sample type	Type of sample, e.g. Serum, Plasma, Tissue —optional
Project / Study	Associated project —optional
Patient/Animal/Trial ID	Subject identification —optional
Collection date	Date of sample collection —optional
Storage date	Pre-filled with today's date
Volume / Amount	e.g. 500 µl, 2 pieces —optional
Status	Available / Removed / Consumed / Reserved / Disposed
Remarks	Free text —optional

No field is mandatory except the internal system ID which is assigned automatically.

4.2 Assigning a Storage Location

All location fields are optional. You can save a sample with no location —it will appear under "Not stored" in the sidebar. You can assign a storage unit without a box; in this case the Storage Note field is available for a free-text description.

For grid boxes a visual raster appears. Occupied positions are marked. For freeform boxes the Storage Note field is always available.

- Ctrl+Click (or Cmd+Click on Mac) to select multiple positions in a grid box. A confirmation prompt will ask before creating one sample per position, all sharing the same metadata.

SAMPLE DATA

Sample ID (optional)

BB-2025-001

Sample type

Serum

Project / Study

Biobank

Patient / Animal / Subject ID

BK-001

Collection date

01-09-2025

Storage date

02-09-2025

Volume / Amount

1 ml

Status *

Available

Remarks

General remarks about the sample

STORAGE LOCATION

Storage Unit

-80er 1

Compartment

Fach 1

Rack

Rack B

Drawer

Schublade 1

Box

Box 4 (8x12)

Grid position *

☐ Free ☒ Occupied

A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12
C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12
E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	E11	E12
F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12

Selected: A1
Ctrl+Click for multi-select

Storage note

Cancel Save

4.3 Autocomplete

All text fields in the sample form and storage structure manager offer autocomplete based on existing values. Each field only suggests values from its own category.

4.4 Edit, Move, Delete

All three actions are available in the detail panel. Move reassigns the storage location without touching other fields. Deleting a sample is permanent —a confirmation is shown first.

5. Managing Storage Structure

5.1 Creating Elements

Open Storage. Each level has its own tab. Storage units have an additional Location field shown in the sidebar and detail panel.

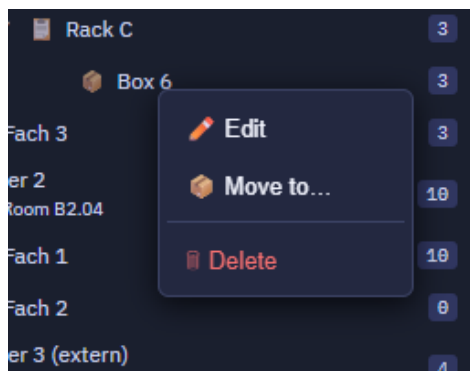
5.2 Groups

The Groups tab lets you create visual groups (e.g. "-20°C devices", "4°C devices"). Assign any storage unit to a group via the dropdown in the same tab. Groups appear as ■ entries in the sidebar tree—they are for display only and do not affect storage paths in any way.

5.3 Right-Click Context Menu

Right-clicking any item in the sidebar opens a context menu with three options:

- Edit —opens a modal for name, location (freezer units), and size (grid/numbered boxes)
- Move to...—moves the element including all contents and samples to a new location
- Delete —deletes the element; shows a warning with sample count if it contains samples



5.4 Drag & Drop

Elements can be reordered within the same level by drag and drop. A blue bar shows the insertion point. A rack can only be moved within its own compartment.

5.5 Resizing Boxes

Right-click a grid or numbered box → Edit → change the Rows and Columns values. If any samples occupy positions that no longer exist after resizing, a warning is shown with the exact positions. After confirmation those samples remain in the box but lose their position.

5.6 Moving Structure Elements

Via Move to...in the context menu, a compartment, rack, drawer or box can be moved to a different parent. All contained sub-elements and samples move with it. Storage paths of affected samples are updated automatically.

■ ■ Deleting a storage unit removes all contained compartments, racks, drawers, boxes and samples. This cannot be undone. Always create a backup first.

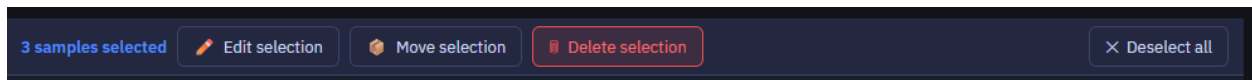
6. Multi-Select and Bulk Actions

Checkboxes on the left of the sample list allow selecting any number of samples. The checkbox in the table header selects all currently visible samples.

Once samples are selected, a bulk action bar appears with three actions:

- Edit selection —update type, status, project or remarks across all selected; only checked fields are overwritten; confirmation before saving
- Move selection —assigns the same new storage location to all selected samples; grid positions are cleared and must be reassigned manually
- Delete selection —deletes all selected samples after confirmation

■ Combine the sidebar tree filter, status filter and search field to quickly select a specific group of samples —e.g. all samples of a project in a specific rack.



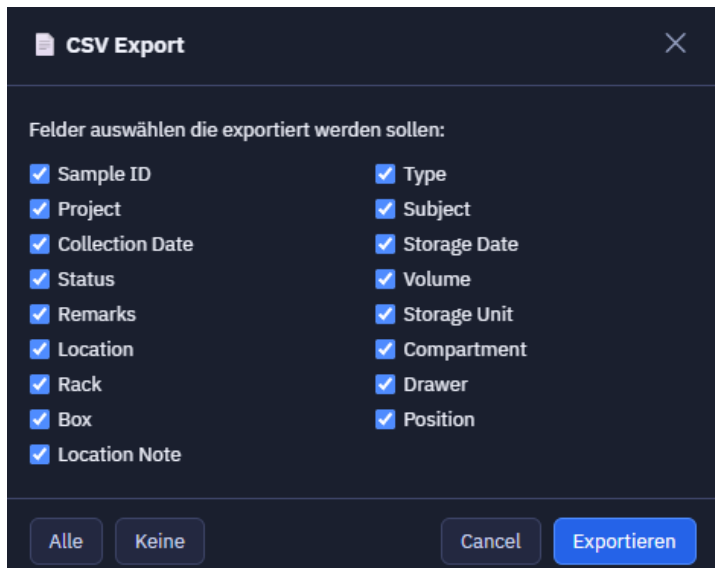
7. Search and Filter

The search field searches all text fields simultaneously: sample ID, system ID, type, project, subject ID, remarks and all location fields. Case-insensitive. The type and status dropdowns provide additional filters. The sidebar tree acts as a third filter—all three work simultaneously.

8. CSV Export and Import

8.1 CSV Export

Click CSV Export to open a field selection dialog with all 17 available fields as checkboxes. Use All / None to quickly toggle. The filename includes today's date automatically. The file uses UTF-8 BOM and semicolon as separator —opens directly in Excel. Column headers switch with the language setting.



8.2 CSV Import

The import workflow has four steps:

- Step 1 —Download the template (CSV with all column headers and 5 example rows). Example rows start with EXAMPLE_ and are automatically skipped on import.
- Step 2 —Fill in the template and upload it via Select file
- Step 3 —Preview: shows a table of the first 10 rows, skipped example rows, and a list of any missing structure elements with options to configure their type and size
- Step 4 —Choose mode: Append to existing database or New database (replaces all existing samples)

Missing structure elements (compartments, racks, drawers, boxes) are created automatically on import. For boxes, the type is auto-detected from the position format —a letter+number position (A1, B3) suggests a grid box; a number-only position (56, 100) suggests a numbered box. You can change the type and size before confirming.

■ The CSV uses semicolons as separator. Dates can be in DD.MM.YYYY or YYYY-MM-DD format.

9. Saving and Backup

9.1 Save

Click Save or use Ctrl+S. If a file is already open a confirmation prompt appears before overwriting. On first save a file dialog opens.

■■ Saving overwrites the current file directly. Create regular backups before making large structural changes.

9.2 Backup

Always opens a Save As dialog regardless of whether a file is currently open. Use this to create dated copies, e.g. samples-backup-20260519.json.

9.3 File Format

The database is a plain JSON file. In an emergency it can be opened in any text editor. Each sample has an internal `_uid` assigned by the system (never changes) and an optional user-visible id field.

10. Audit Log

The Log button opens the change history. All sample actions are logged: create, edit, move, delete, and bulk actions. Each entry includes a timestamp and description. The log is stored in the JSON file and cannot be edited through the interface.

11. Practical Examples

Example 1: First Setup

Lab with two -80°C freezers in Room 3.01 and an external provider:

- Storage → Groups tab → add group "-80°C freezers"
- Storage → Storage units → add "-80°C #1" with location "Room 3.01"
- Add "-80°C extern" with location "Kryolab GmbH, Building 7"
- Assign both units to the "-80°C freezers" group
- Add compartments, racks, drawers and boxes as needed
- First sample: + New Sample, leave ID empty, select type "Serum", choose box and position

Example 2: Emergency Storage Without Full Structure

A sample must be stored immediately before the box is set up:

- + New Sample, leave ID empty, enter type and project
- Select the storage unit and rack, leave box empty
- In the Storage Note field: "Drawer 2, front left, blue bag"
- Later edit the sample to add the box once it has been created

Example 3: Batch Status Change

After a study ends, 20 samples need to be set to "Consumed":

- Select the relevant rack in the sidebar
- Filter: Project = Study Alpha
- Activate all checkboxes (header checkbox selects all visible)
- Edit selection → check Status → select Consumed → Save

Example 4: Import from Excel

Importing a list of 200 reagents from an existing spreadsheet:

- Download the import template from CSV Import
- Copy your data into the template columns —leave EXAMPLE_ rows or delete them
- Upload the filled template
- Review the preview —missing boxes appear with auto-detected type
- Adjust box type/size if needed, choose Append, click Import

Data Backup

- Store the JSON file on a network drive with automatic versioning
- Create weekly backups with a date in the filename using the Backup function
- Always back up before major restructuring

Multiple Users

Locus Arca is designed as a single-user system. Clarify internally who has write access at any given time. Simultaneous editing of the same JSON file can cause data loss.

No Admin Rights Required

The application runs as a single HTML file opened directly in the browser. No installation, no elevated permissions needed. It can be used on locked-down workstations typical in hospitals and research institutions.

Browser Compatibility

Browser	Direct save to file
Chrome	✓ Yes (recommended)
Edge	✓ Yes (recommended)
Opera / Brave / Vivaldi	✓ Yes (Chromium-based)
Firefox	■ Download only (fully functional)
Safari	■ Download only (fully functional)